MACHINE LEARNING TECHNICAL PRESENTATION

Iris Flower Classification Using Decision Trees

An end-to-end evaluation of decision boundaries, feature selection, and predictive performance.

Executive Overview

High Predictive Precision

96.67% overall test set accuracy achieved.

Controlled Complexity

Strict depth cap of 3 prevents overfitting.

Interpretive Rules

Clear petal measurement thresholds derived.

Test Accuracy

96.67%

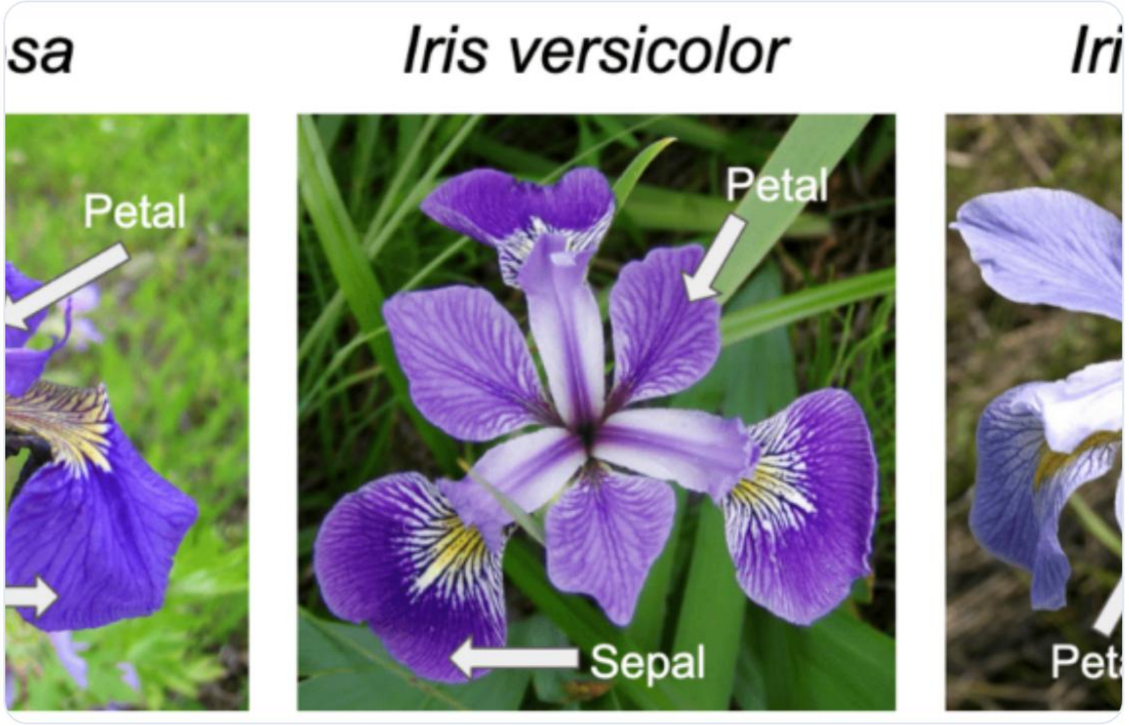
Model generalization evaluated on holdout test dataset

Iris Dataset Characteristics

150 Total Samples
Zero missing or null values in data.

3 Target Classes
Setosa, Versicolor, and Virginica species.

4 Morphological Features
Sepal length/width & petal length/width.



Execution Pipeline Architecture

STEP 01

Ingestion

Data loaded using Pandas and Glob search.

STEP 02

Stratified Split

80% train and 20% test partition ratio.

STEP 03

Tree Fitting

Trained with Gini Impurity metric.

STEP 04

Evaluation

Validated with confusion matrix metrics.

Classification Performance Report

Species Class	Precision	Recall	F1-Score	Support
Iris-setosa	1.00	1.00	**1.00**	10
Iris-versicolor	1.00	0.90	**0.95**	10
Iris-virginica	0.91	1.00	**0.95**	10

Decision Logic Structure

Root Node: Petal Length ≤ 2.45 cm

IF TRUE

Iris-setosa (Pure Node)

IF FALSE

Check Petal Width ≤ 1.75 cm

Key Model Insights

Dominant Predictor

Petal dimensions contain maximum split information.

Linear Separability

Iris-setosa isolates completely at first decision split.

Optimal Regularization

Max depth of 3 prevents variance while holding 96.67% accuracy.

Questions & Answers

Iris Decision Tree Classification Technical Presentation

Image Sources



https://camo.githubusercontent.com/ca4ff60b76c0803d41de1ebf12d6617465e3924fe6845192dd98e9f8503596c2/68747470733a2f2f7777772e656d6265646465642d726f626f746963732e636f6d2f77702d636f6e74656e742f75706c6f6164732f323032322f30312f497269732d446174617365742d436c617373696669636174696f6e2d313032
34783336372e706e67

Source: github.com